

Hugo Mirault

Augusta USA

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Research Interest

Distributed algorithms, Information theory, Concurrency, Fault resilience, Byzantine resilience, Distributed protocols.

Education

PhD in Computer Science: 2024-

Augusta University (USA)

PhD in Distributed Algorithms under the supervision of Dr. Robinson.

Master's in Algorithmics (ALGO/MIT): 2020-2022 (With Honors)

University of Montpellier

Algebraic Computation, Cryptography, Constraint Programming, Graph Theory, Complexity Theory, Information Theory, Computational Models, Operations Research, Formal Methods.

Bachelor's Degree in Computer Science: 2017-2020 (With Honors)

University of Montpellier

Bachelor's Degree in General Computer Science

Publications [See Google Scholar for more](#)

Time-Optimal and Energy-Efficient Deterministic Consensus

OPODIS 2025

Shachar Meir, Hugo Mirault, David Peleg, and Peter Robinson. In Proceedings of the 29th International Conference on Principles of Distributed Systems (OPODIS 2025) <https://doi.org/10.4230/LIPIcs.OPODIS.2025.15>

Brief Announcement: Towards Energy-Efficient Distributed Agreement

PODC 2025

Hugo Mirault and Peter Robinson. In Proceedings of the ACM Symposium on Principles of Distributed Computing. Association for Computing Machinery. <https://doi.org/10.1145/3732772.3733554>

Brief Announcement: Perfect Matching with Few Link Activations

SIROCCO 2025

Mirault, H., Robinson, P., Tan, M.M., Zhu, X. Brief Announcement: In Structural Information and Communication Complexity. Lecture Notes in Computer Science, vol 15671. Springer, Cham. https://doi.org/10.1007/978-3-031-91736-3_28

Professional experiences

Graduate Research Assistant: 2024-2027

Augusta University

Graduate Research Assistant under supervision of Dr. Robinson.

Formal Methods Engineer: 2023

Systemrel

Formal Methods Engineer for the Design/Formal Verification of Automatic Trains (CBTC). B Method and Interactive Proof.

End-of-study internship - Secure multiparty computation: 2022

LIRMM

Study of secure multiparty computation protocols (MPC), in the context of algebraic computation on matrices with polynomial coefficients. Writing of the final dissertation. Final grade: 15/20. <https://doi.org/10.48550/arXiv.2211.06732>

Student Project - Scheduler: 2021

University of Montpellier

Modeling of a scheduling problem, followed by implementations, development of an exact linear programming (MIP) solver and heuristics. Study and comparison of heuristics with the exact solver. Final grade: 15.63/20

Research Internship - Kolmogorov Complexity and Randomness: 2021

LIRMM

Study of the link between Kolmogorov prefix complexity and the Martin-löf notion of randomness, presented on the Chaitin constant.

Langues

French Mother Tongue

English Fluent (B2/C1)